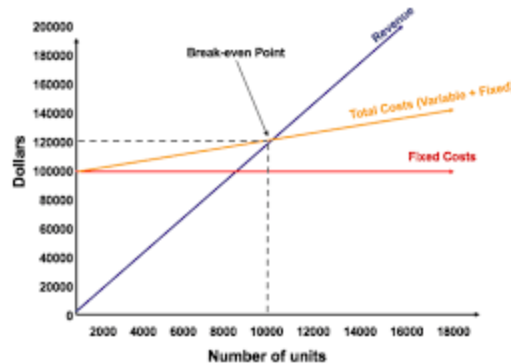


Break-Even Analysis

A break-even analysis calculates the point at which total revenue equals total costs (zero profit or loss), helping businesses determine the necessary sales volume, price, or cost structure to become profitable.



The formula is: **Break-Even Units = Total Fixed Costs / (Selling Price per Unit - Variable Cost per Unit).**

Break Even Formula Variations:

- Break Even Points (Units) = Total Fixed Costs / Contribution Margin
- Contribution Margin (Units) = Selling Price per Unit – Variable Cost per Unit
- Break Even Points (Sales Dollars) = Total Fixed Costs / Contribution Margin Ratio
- Contribution Margin Ratio = Contribution Margin / Sales Price

Example-1 (Manufacturing)

A manufacturer has fixed monthly costs (rent, insurance, salaries) of \$6,000. It costs \$2 in raw material and packaging (variable cost) to make each unit, which sells for \$7 (selling price).

How many units to sell for break-even?

1. Contribution Margin = \$7(Price) - \$2(Variable Cost) = \$5 (Revenue per unit to cover fixed costs)
2. Break-even Units = \$6,000 (Fixed Cost) / \$5 (Contribution Margin) = 1,200 units.

Conclusion:

The manufacturer must sell 1,200 units monthly to break even.

Selling 1,201 units results in a profit.

Selling 1,199 units results in a loss.

Example 2: SaaS Product Launch

A tech company is launching a new SaaS feature. They need to know how many subscriptions are required to break even.

- Fixed Costs (Development, Marketing, Onboarding): \$50,000
- Selling Price per Subscription: \$40
- Variable Cost per Subscription (Support, Hosting): \$15

1. Calculate Contribution Margin per Unit:

Price – Variable Cost = \$40-\$15 = \$25

(Every subscription contributes \$25 toward covering the fixed costs.)

2. Calculate Break-Even Point (Units):

Fixed Costs / Contribution Margin = \$50,000 / \$25 = 2,000 subscriptions

(The Company must sell 2,000 subscriptions to break even.)

3. Impact of Changes:

- If Fixed Costs rise to \$60,000, the break-even point increases to 2,400 units (\$60,000 / \$25).
- If Price drops to \$35 (new margin \$20), the break-even point rises to 2,500 units (\$50,000 / \$20).

Key Metrics for Software Projects:

- **Fixed Costs:** Development salaries, infrastructure setup, marketing campaigns, and software licenses.
- **Variable Costs:** Cloud hosting usage (AWS/Azure), customer support hours, payment processing fees.
- **Margin of Safety:** The gap between expected sales and the break-even point; a larger gap implies lower risk.